Verification of Open Interactive Markov Chains: A Comprehensive Overview

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1 Introduction

Interactive Markov Chains (IMCs) blend the non-determinism of labeled transition systems (LTS) with the stochastic timing of continuous-time Markov chains (CTMCs). They are pivotal for modeling systems where both controlled decisions (e.g., component interactions) and random events (e.g., failures) coexist. This paper addresses **open systems** where components interact with unknown environments.

Key Motivation:

- **Component-Based Systems**: Modern systems (e.g., autonomous vehicles, cloud infrastructure) rely on third-party components with uncertain behavior.
- **Dependability Guarantees**: Ensure performance (e.g., response time) and reliability (e.g., fault tolerance) even when parts of the system are unspecified.

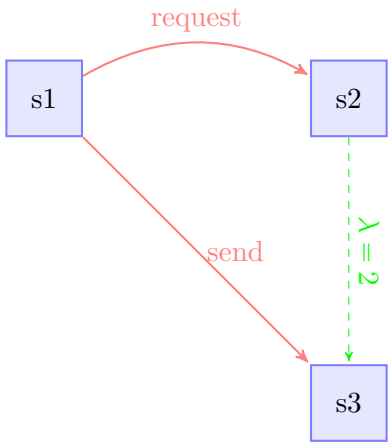
Example 1. *A fault-tolerant workstation cluster with redundant components. The goal is to compute the worst-case probability of maintaining "premium service" (e.g., $\geqslant N$ operational nodes) within time T, even if some components (e.g., switches) are left unspecified.*

2 Interactive Markov Chains (IMCs): Formal Definition

An IMC is a tuple $\mathcal{C} = (S, \mathrm{Act}^\tau, \hookrightarrow, \rightsquigarrow, s_0)$:

- **States**: Finite set S.
- **Actions**: Act^τ includes internal (τ) and external actions.
- **Interactive Transitions**: $s \overset{a}{\hookrightarrow} s'$ for $a \in \mathrm{Act}^\tau$.
- **Markovian Transitions**: $s \overset{\lambda}{\rightsquigarrow} s'$ with rate λ.

We also define a total exit rate function $E : S \to \mathbb{R}_{\geqslant 0}$ which assigns to each state the sum of rates of all outgoing Markovian transitions, i.e., $E(s) = \sum_{t \in \mathrm{succ}_M(s)} \lambda(s,t)$, with the convention that the sum is zero if $\mathrm{succ}_M(s)$ is empty. Furthermore, we define a probability matrix $P(s,t)$ by $P(s,t) = \begin{cases} \dfrac{\lambda(s,t)}{E(s)} & \text{if } s \overset{\lambda}{\rightsquigarrow} t, \\ 0 & \text{otherwise.} \end{cases}$.



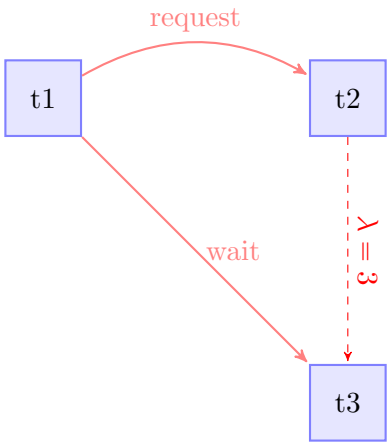


Figure 1: Two different IMCs part of the same system. Ie. component IMCs of a system.

3 Compositional Operators

3.1 Parallel Composition ($\|_A$)

Synchronizes components on shared actions A. Non-synchronized actions interleave. Given **IMCs**

$$\mathcal{C}_1 = (S_1, \mathbb{A}\mathrm{ct}_1^\tau, \hookrightarrow_1, \rightsquigarrow_1, s_{01}) \text{ and } \mathcal{C}_2 = (S_2, \mathbb{A}\mathrm{ct}_2^\tau, \hookrightarrow_2, \rightsquigarrow_2, s_{02}),$$

and **synchronisation alphabet** $A \subseteq \mathbb{A}ct_1 \cap \mathbb{A}ct_2$, their parallel composition $\mathcal{C}_1 \,||_A\, \mathcal{C}_2$ is given as

$$\mathcal{C} = (S_1 \times S_2, \mathbb{A}\mathrm{ct}_1^\tau \times \mathbb{A}\mathrm{ct}_2^\tau, \hookrightarrow, \rightsquigarrow, (s_{01}, s_{02}))$$

NOTE:

- $s_1 \overset{a}{\hookrightarrow} s_1'$ and $s_2 \overset{a}{\hookrightarrow} s_2'$ and $a \in A$ implies $(s_1, s_2) \overset{a}{\hookrightarrow} (s_1', s_2')$
- $s_1 \overset{a}{\hookrightarrow} s_1'$ and $a \notin A$ implies $(s_1, s_2) \overset{a}{\hookrightarrow} (s_1', s_2)$ for each $s_2 \in S_2$
- $s_2 \overset{a}{\hookrightarrow} s_2'$ and $a \notin A$ implies $(s_1, s_2) \overset{a}{\hookrightarrow} (s_1, s_2')$ for each $s_1 \in S_1$
- $s_1 \overset{\lambda}{\rightarrow} s_1'$ implies $(s_1, s_2) \overset{\lambda}{\rightsquigarrow} (s_1', s_2)$ for each $s_1 \in S_1$
- $s_2 \overset{\lambda}{\rightarrow} s_2'$ implies $(s_1, s_2) \overset{\lambda}{\rightsquigarrow} (s_1, s_2')$ for each $s_2 \in S_2$

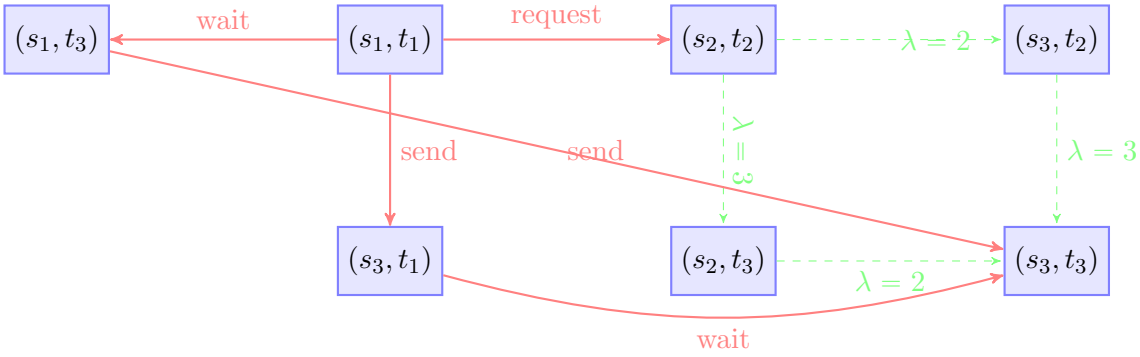


Figure 2: Parallel composition of two IMCs. Synchronized on *request* action, with independent *send/wait* actions and combined Markovian transitions.

3.2 Hiding ($\backslash A$)

Converts external actions A to internal τ, closing the system. Given IMC $\mathcal{C} = (S, \mathbb{A}\mathrm{ct}^\tau, \hookrightarrow, \rightsquigarrow, s_0)$, a hidden alphabet $A \subseteq \mathbb{A}ct$, the hiding $\mathcal{C}\backslash A$ is $\mathcal{C}' = (S, \mathbb{A}\mathrm{ct}^\tau \setminus A, \hookrightarrow', \rightsquigarrow, s_0)$ where $\hookrightarrow'$ is the smallest relation satisfying

- $a \in A$ implies $s \overset{\tau}{\hookrightarrow} s'$, and,
- $a \notin A$ $s \overset{a}{\hookrightarrow}' s'$

for each $s \overset{a}{\hookrightarrow} s'$

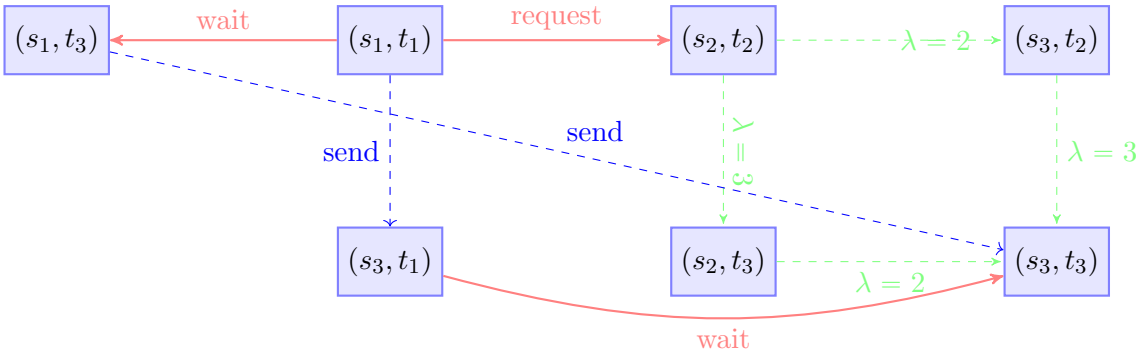


Figure 3: The *send* actions have been hidden, reflecting internal behavior not observable in the composed model. This allows the user to get a simplified view. If user sends wait then you reach state $(s3, t3)$.

3.3 Scheduler

A *scheduler* for an IMC $C = \left(S, \mathsf{Act}_\tau, \rightarrow, \rightsquigarrow, s_0\right)$ is a measurable function $\sigma : (S \times \mathbb{R}_{\geqslant 0})^* \times S \to S$ such that for each history $h = s_0\, t_0\, s_1 \cdots s_n$ with $\mathrm{succ}_\tau(s_n) \neq \varnothing$, we have $\sigma(h) \in \mathrm{succ}_\tau(s_n)$. The set of all schedulers for C is denoted by $\mathcal{S}(C)$. The decision of the scheduler $\sigma(h)$ determines t_n and s_{n+1} as follows:

1. **Case 1:** If $\mathrm{succ}_\tau(s_n) \neq \varnothing$, then the run proceeds immediately. That is, the time is set to $t_n := 0$, and the next state is given by $s_{n+1} := \sigma(h)$.

2. **Case 2:** Otherwise, if $\mathrm{succ}_\tau(s_n) = \varnothing$, then only Markovian transitions are available in s_n. In this case, the run moves to a randomly chosen next state s_{n+1} with probability $P(s_n, s_{n+1})$, after waiting for a random time t_n, which is chosen according to the exponential distribution with rate $E(s_n)$.

4 Compositional Verification

Assumption 1. *Each cycle contains a Markovian* transition.
It implies that the probability of taking infinitely many transitions in finite time, i.e., of Zeno behavior, is zero. This assumed property does not affect power of model and is a compositional one, i.e., it is preserved by parallel composition and hiding.

Assumption 2. *Internal and external actions are not enabled at the same time, i.e., for each state s, either* $\mathrm{succe}(s) = \varnothing$ *or* $\mathrm{succ}_\tau(s) = \varnothing$.
Assumption 2 is not compositional; specifically, it is not preserved by applications of the hiding operator.
For a given environment E with the same $\mathbb{A}\mathrm{ct}^\tau$, we give the composition:

$$\mathcal{C}(E) \;=\; (\mathcal{C} \;||_{\mathbb{A}ct}\; E)\backslash \mathbb{A}ct$$

Consider σ scheduler of $\mathcal{C}$ and π a scheduler of $\mathcal{C}(E)$; π respects σ if for every history $\mathfrak{h}$ of $\mathcal{C}(E)$:

- $\pi(\mathfrak{h}) = (c, e)$ where $c_n \overset{a}{\hookrightarrow} c$ and $e_n \overset{a}{\hookrightarrow} e$ (synchronisation resolved)
- $\pi(\mathfrak{h}) = (c_n, e)$ where $e_n \overset{\tau}{\hookrightarrow} e$ (π governs choice of moves in E)
- $\pi(\mathfrak{h}) = (\sigma(\mathfrak{h}_\mathcal{C}), e_n)$ where $e_n \overset{\tau}{\hookrightarrow} e$ where $h_\mathcal{C}$ is $\mathfrak{h}$ projecting over the states of E (π chooses moves in $\mathcal{C}$ as per σ)

In $\mathcal{C}(E)$, the value of time-bounded reachability is

$$R^{\mathcal{C}(E)}_{T,G} = \sup_{\sigma \in \mathfrak{S}(\mathcal{C})} \inf_{\substack{E \in \mathrm{ENV} \\ \pi \in \mathfrak{S}(\mathcal{C}(E))}} P^\pi_{\mathcal{C}(E)}[\Diamond^{\leqslant T} G_E] \tag{1}$$

where $G_E = G \times S_E$ and ENV is the set of all IMCs with alphabet $\mathbb{A}\mathrm{ct}$

Theorem 3.1. Let $\varepsilon > 0$ be an approximation bound and $C = (S, \mathsf{Act}_\tau, \to, \rightsquigarrow, s_0)$ be an IMC satisfying Assumptions 1 and 2. Then one can approximate the value of compositional time-bounded reachability of C up to ε and compute an ε-optimal scheduler in time $O\big(|S|^2 \cdot (\lambda T)^2/\varepsilon\big)$, where λ is the maximal rate of C and T is the reachability time-bound.

5 Game of Environment and Controller

In order to approximate $R_{T,G}^{\mathcal{C}(E)}$, the value of compositional time-bounded reachability, we transform the IMC C into a two-player controller–environment (CE) game G.
The game G is played on the graph of the IMC C by two players: con (controlling the component C) and env (simulating the environment). Essentially, in states with internal transitions, con selects one transition; in states with external (synchronizing) transitions, env either selects one of these transitions or chooses a delay $t_e \in \mathbb{R}_{>0}$.
By Assumption 2, the players act on disjoint sets of states, making G a turn-based game. Both internal and external transitions take zero time when executed. If no zero-time transition is chosen, the delay t_e competes with a Markovian transition, which is executed after a random time t_M sampled from the exponential distribution with rate $E(s)$.
Here, the objective of con is to reach a target set of states G within a time bound T, while env aims to prevent this.

Formally, let the IMC be given by

$$C = (S, \mathsf{Act}_\tau, \rightarrow, \rightsquigarrow, s_0),$$

and consider the corresponding CE game G.
A run of G is an infinite sequence

$$s_0\, t_0\, s_1\, t_1\, s_2\, t_2\, \ldots,$$

where $s_n \in S$ is the n-th state and $t_n \in \mathbb{R}_{\geqslant 0}$ is the time spent there. Based on the history $s_0\, t_0\, \ldots\, t_{n-1}\, s_n$, the players choose moves as follows:

1. If $\mathrm{succ}_\tau(s_n) \neq \varnothing$, then con chooses a state $s_\tau \in \mathrm{succ}_\tau(s_n)$.
2. Otherwise, env chooses either a state $s_e \in \mathrm{succe}(s_n)$ or a delay $t_e \in \mathbb{R}_{>0}$ (if $\mathrm{succe}(s_n) = \varnothing$, only a delay can be chosen).

Next, if Markovian transitions are available, they are resolved by randomly choosing a target state s_M according to the distribution $P(s_n, \cdot)$ and sampling a time t_M from the exponential distribution with rate $E(s_n)$. The next waiting time t_n and state s_{n+1} are determined as follows:

1. If $\mathrm{succ}_\tau(s_n) \neq \varnothing$ and s_τ is chosen, then $t_n = 0$ and $s_{n+1} = s_\tau$.
2. If s_e is chosen, then $t_n = 0$ and $s_{n+1} = s_e$.
3. If a delay t_e is chosen:
   - If $\mathrm{succ}_M(s_n) = \varnothing$, then $t_n = t_e$ and $s_{n+1} = s_n$.
   - If $t_e \leqslant t_M$, then $t_n = t_e$ and $s_{n+1} = s_n$.
   - If $t_M < t_e$, then $t_n = t_M$ and $s_{n+1} = s_M$.

$\sigma : (S \times \mathbb{R}_{\geqslant 0})^* \times S \to S$: Taken as strategy of con, and
$\pi : (S \times \mathbb{R}_{\geqslant 0})^* \times S \to S \cup \mathbb{R}_{>0}$: Taken as a strategy of env.

We denote by Σ and Π the sets of strategies for con and env, respectively. To prevent Zeno behavior, we restrict Π to strategies for which the sum of chosen delays diverges almost surely, regardless of con's moves.

Value of reachability in this game, $\mathcal{G}$ is given by

$$R^{\mathcal{G}}_{T,G} = \sup_{\sigma\in\Sigma}\inf_{\pi\in\Pi}\ \mathcal{P}^{\sigma,\pi}_{\mathcal{G}}[\Diamond^{\leqslant T} G] \tag{2}$$

Proposition: $R^{\mathcal{C}(E)}_{T,G} = R^{\mathcal{G}}_{T,G}$
That is, we need to show

$$\sup_{\sigma\in S(C)}\ \inf_{\substack{E\in ENV\\ \pi\in S(C(E),\sigma)}} P^{\pi}_{C(E)}\left[\Diamond^{\leqslant T}_{G_E}\right] = \sup_{\sigma\in\Sigma}\inf_{\pi\in\Pi} P^{\sigma,\pi}_{G}\left[\Diamond^{\leqslant T}_{G}\right]$$

We start by considering the inequality (1) $\geqslant$ (2). Pick any scheduler $\sigma \in \Sigma$ and fix an environment E along with a scheduler $\pi \in S(C(E), \sigma)$. The key idea is that the environment E (under π) is responsible for choosing the delays before external actions occur. These delays are determined by a series of internal and Markovian transitions that occur before an external action is taken —similar to the role played by env in the CE game. The difference is that E selects these delays randomly, while env uses a fixed (deterministic) strategy. Using a more detailed argument, we can eliminate this randomness and obtain a strategy π' in the CE game that satisfies: $P^{\sigma,\pi'}_{G}\left[\Diamond^{\leqslant T}_{G}\right] \leqslant P^{\pi}_{C(E)}\left[\Diamond^{\leqslant T}_{G_E}\right]$.

Concerning the second inequality (1) $\leqslant$ (2), we show that every strategy of env can be (approximately) implemented using a suitable environment together with a scheduler π. The idea is to simulate every deterministic delay, say t, chosen by env using a random delay tightly concentrated around t (roughly corresponding to an Erlang distribution) that is implemented as an IMC. We show that the imprecision of delays introduced by this randomization induces only negligible alteration to the value.

6 Discretization

We now try to convert the CE game into a discrete time-based stochastic game with time proceeding by a fixed time step $\delta = T/N$ where $N \in \mathbb{N}$. The discrete stochastic game Δ is played on a graph whose set of vertices V such that $V = V_{\square} \uplus V_{\Diamond} \uplus V_{\circ}$ where **con** $\rightarrow \square$, **env** $\rightarrow \Diamond$ and a randomly choosing agent $\circ$

Δ is constructed by replacing the vertices form $\mathcal{G}$ as shown in the figures in the following page. The *value* of Δ, as defined for IMCs and CE games, is defined as

$$R^{\Delta}_{T,G} = \sup_{\sigma\in\Sigma_\Delta}\inf_{\pi\in\Pi_\Delta}\ \mathcal{P}^{\sigma,\pi}_{\Delta}[\Diamond^{\#_\circ\leqslant N} G]$$

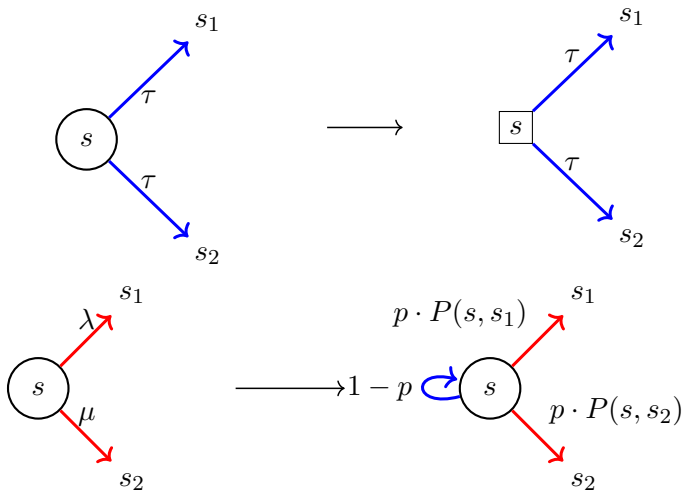


Figure 4: Conversion of vertices given that they have (a) only internal transitions, (b) only Markovian transitions

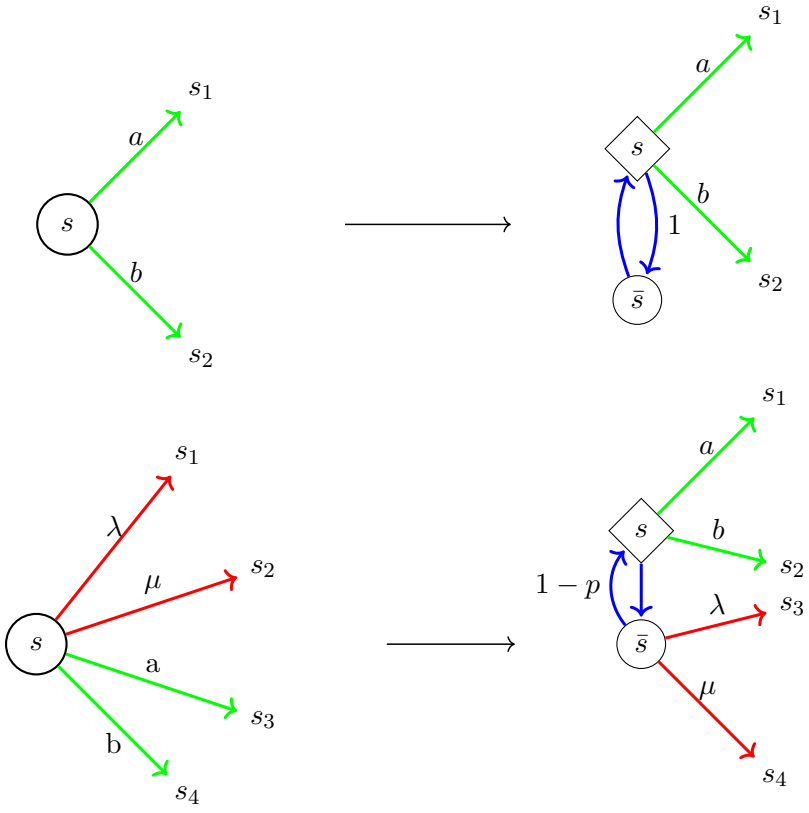


Figure 5: Conversion of vertices given that they have (a) only external transitions, (b) both external and Markovian transitions

A strategy is *counting* if only the last vertex and the current count $\#_\circ$; can be represented as a fxn : $V \times \{0, ...N\} \to V$. One of the most important lemmas presented by the paper is

There are counting strategies optimal in $R^{\Delta}_{T,G}$ and they can be computed with $R^{\Delta}_{T,G}$ in $\mathcal{O}(N|V|^2)$

A crucial question raised during the presentation was how do we show that this result for counting strategies in discretized games can be extended to optimal strategies in the IMCs. For this we define grid strategies:

We say that a strategy is a *grid strategy* on a grid of size $\delta > 0$ if

- it decides only according to the current state and the integer k such that the total time of the history belongs to the interval $[k\delta, (k+1)\delta)$, i.e. for any two histories $h = s_0 t_0 \dots t_{n-1} s_n$ and $h' = s'_0 t'_0 \dots t'_{m-1} s'_m$ with $s_n = s'_m$ and $\sum h, \sum h' \in [k\delta, (k+1)\delta)$ for some $k \in \mathbb{N}_0$ we have $\sigma(h) = \sigma(h')$; and
- it is either a strategy of the player **con** or it chooses waiting times only on the δ-grid, i.e. for any history $h = s_0 t_0 \dots t_n s_n$ the strategy σ either chooses an action or a time step t_{n+1} such that $\sum h + t_{n+1} = k\delta$ for some $k \in \mathbb{N}$.

Furthermore, for $\lambda \in \mathbb{R}$ we say that a strategy π of the player *envis exponential* with rate λ if

- it chooses to wait solely with the exponential distribution with rate λ;
- for any history $h = s_0 t_0 \dots t_n s_n$ with $n > 0$, $t_n > 0$, and $s_n \neq s_{n-1}$ it chooses to wait;
- for other histories it behaves as a grid strategy for some $\delta > 0$, i.e. we have $\sigma(h) = \sigma(h')$ for any two histories $h = s_0 t_0 \dots t_{n-1} s_n$ and $h' = s'_0 t'_0 \dots t'_{m-1} s'_m$ with
  - either $n = m = 0$ or $t_{n-1} = t'_{m-1} = 0$ or $s_n = s_{n-1} = s'_m = s'_{m-1}$, and
  - $s_n = s'_m$ and $\sum h, \sum h' \in [k\delta, (k+1)\delta)$ for some $k \in \mathbb{N}_0$. *[grid strategy].*

In Lemma A.9, the paper provides a structural property that simplifies the analysis of strategies in the given framework. Intuitively, it ensures that certain constraints on waiting times or transitions can be enforced without loss of generality. This helps in reducing the complexity of decision-making by allowing a decomposition into manageable cases, often leveraging grid-based or exponential timing strategies. This shows that considering grid strategies and by extension, counting strategies, for the discretized games is sufficient to construct the optimal scheduler for the corresponding IMC. For the detailed proof, please refer to the reference.

7 Conclusion & Further Analysis

Through this paper, we looked at the computation of maximal timed bounded reachability for IMC operating in an unknown IMC environment. We first showed that the resulting stochastic two-player game has the same extremal values as a CE-game. The value of

the CE-game was then approximated up to a given precision by discretization and the resulting control strategy translated back to a scheduler of the IMC achieving the bound.

This paper analysed the time-bounded reachability for a given IMC, which leads to the obvious question of what about time unbounded reachability? Of course, this has also been a well studied question and is covered in depth in the reference 2 given. The paper shows that Total-time deterministic positional policies are optimal for maximising the reachability probability. Using this result along with the concept of discretization as discussed in this paper itself, a polynomial time algorithm can be developed to calculate the maximal reachability probability given an IMC and its goal states.

8 References

1. Verification of Open Interactive Markov Chains [2012], Authored by: Tomáš Brázdil, Holger Hermanns, Jan Krčál, Jan Křetínský and Vojtěch Řehák

2. The How and Why of Interactive Markov Chains, Authored by: Holger Hermanns and Joost-Pieter Katoen

3. Model checking nondeterministic and randomly timed systems, by M.R. Neuhäußer